Power Flow Analysis

Lecture Notes

1 Power Equation

$$[I] = [Y][V], \qquad I_i = \sum_{j=1}^{n} Y_{ij} V_j$$

(e.g. for $n = 5$: $j = 1, 2, 3, 4, 5$; for $i = 2$ this gives I_2.)

Since $S_i^* = P_i - jQ_i = V_i^* I_i$:

$$\frac{P_i - jQ_i}{V_i^*} = \sum_{j=1}^{n} Y_{ij} V_j$$

Separating the $j = i$ term:

$$\frac{P_i - jQ_i}{V_i^*} = Y_{ii} V_i + \sum_{\substack{j=1 \\ j \neq i}}^{n} Y_{ij} V_j$$

Solving for V_i gives the iterative (Gauss) update:

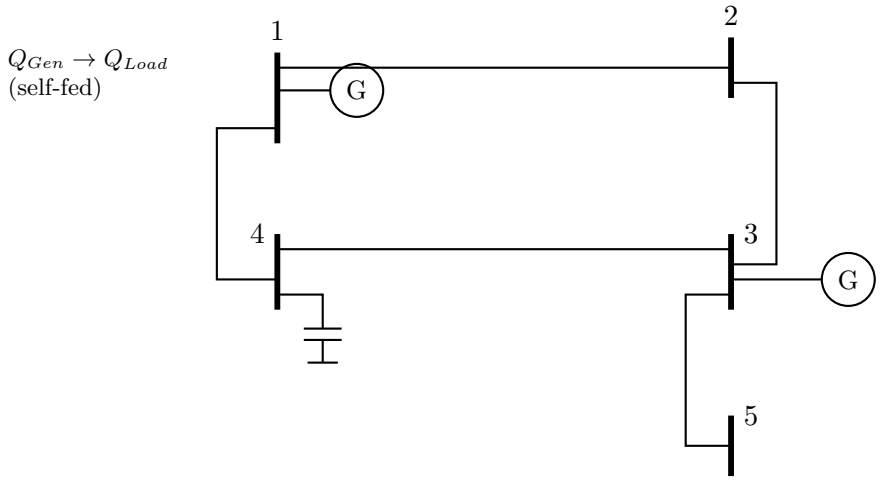
$$V_i^{k+1} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{(V_i^*)^k} - \sum_{\substack{j=1 \\ j \neq i}}^{n} Y_{ij} V_j^k \right]$$

For $n = 5$:

$$V_1 = \frac{1}{Y_{11}} \left[\frac{P_1 - jQ_1}{V_1^*} - \sum_{\substack{j=1 \\ j \neq 1}}^{n} Y_{1j} V_j \right], \qquad V_2 = \cdots, \quad V_3 = \cdots, \quad V_4 = \cdots, \quad V_5 = \cdots$$

2 5-Bus System Example

Bus No.	Type	$\mathbf{P_{Gen}}$	$\mathbf{Q_{Gen}}$	$\mathbf{P_{Load}}$	$\mathbf{Q_{Load}}$	$\mathbf{\lvert V \rvert}$
1	Gen	P_{Gen1}	–	–	–	1.0 p.u.
2	Load	–	–	✓	✓	–
3	Gen	✓	–	–	–	✓
4	Load	–	–	✓	✓	–
5	Load	–	–	✓	✓	–



Voltage drop along a line, e.g. bus 1 → bus 2: $V_2 = V_1 - IZ$.

Update equation (all buses)

$$V_i^{k+1} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{(V_i^*)^k} - \sum_{\substack{j=1 \\ j \neq i}}^{n} Y_{ij} V_j^k \right]$$

Operating limits

$$0.95 \text{ p.u.} \leq V_i \leq 1.05 \text{ p.u.}$$
$$0.9 \text{ p.u.} \leq V_i \leq 1.1 \text{ p.u.} \qquad \text{(e.g. 0.89 p.u. would be a violation)}$$
$$S_{ij} \leq S_{ij,\max} \qquad \text{(e.g. } S_{ij} \leq 500 \text{ MVA)}$$

Initial voltage guess: $V = 1\angle 0^\circ$

3 Gauss Method – Algorithm

1) Read data, form Y_{Bus}, assume initial bus voltage magnitudes and angles. Set $k = 0$.
2) Find V_i^{k+1} for all buses.
3) If $\max_i |V_i^{k+1} - V_i^k| < 0.001 \;\Rightarrow$ **stop**.
   Else, with $\Delta V_i^k = V_i^{k+1} - V_i^k$: $V_i^{k+1} = V_i^k + \Delta V_i^k$; set $k = k + 1$ and repeat step 2.

In the Gauss method, V_i^{k+1} is computed using only the old values $V_1^k, V_2^k, V_3^k, V_4^k, V_5^k$.

3.1 Modification → Gauss–Seidel Method

Use the newest available values as soon as they are computed:

$$V_1^{k+1} \leftarrow V_1^k,\ V_2^k,\ V_3^k,\ V_4^k,\ V_5^k$$
$$V_2^{k+1} \leftarrow V_1^{k+1},\ V_2^k,\ V_3^k,\ V_4^k,\ V_5^k$$
$$V_3^{k+1} \leftarrow V_1^{k+1},\ V_2^{k+1},\ V_3^k,\ V_4^k,\ V_5^k$$

Acceleration factor:

$$V_i^{k+1} = V_i^k + \alpha\, \Delta V_i^k, \qquad \alpha \approx 1.7 \text{ to } 2.0$$

4 Defining a Swing / Slack / Reference Bus

Let us assume Gen 1 as the reference.

- $|V_1| = |V_{set}|$ ← decided by the **Optimal Power Flow (OPF)**.
- Based on load–generation scenarios (balancing territory), OPF sets all other control parameters, i.e. setting of generator bus voltages, taps, shunts, PE-based controllers, etc.

$$\begin{aligned} &\text{Gen 1:} \quad |V_1|\angle 0^\circ = |V_{1,set}|\angle 0^\circ \quad \leftarrow \text{Ref.} \\ &\text{Gen 3:} \quad |V_3|\angle \delta_3 = |V_{3,set}|\angle \delta_3 \quad \rightarrow \delta_3 \text{ changes between iterations} \end{aligned}$$

4.1 Synchronous Generator

- Governor control → frequency (f)
- Excitation control → Automatic Voltage Regulation (AVR) → voltage

Reactive power limits: $Q_{Gen,\min} \le Q_{Gen} \le Q_{Gen,\max}$

$Q_{Gen,cal}$ is obtained from $|V|$, δ of the solution:

- If $Q_{Gen,cal} < Q_{Gen,\min}$ (less than minimum) $\Rightarrow$ $Q_{Gen} = Q_{Gen,\min}$
- If $Q_{Gen,cal} > Q_{Gen,\max}$ (greater than maximum) $\Rightarrow$ $Q_{Gen} = Q_{Gen,\max}$

$\Rightarrow$ the voltage will change (it can no longer be held at V_{set}).

Then for the generator: P_{Gen}, Q_{Gen} (hence P_i, Q_i) are *known*; $|V_i|, \delta_i$ are *not known*.

> The PV bus would be treated as a load → **PQ bus**.

$$\begin{aligned} &\text{Gen} \quad \Leftrightarrow \quad \text{PV bus} \quad \Leftrightarrow \quad P \text{ and } |V| \text{ are known} \\ &\text{Load} \quad \Leftrightarrow \quad \text{PQ bus} \quad \Leftrightarrow \quad P \text{ and } Q \text{ are known} \end{aligned}$$

5 Algorithm with Inclusion of PV Buses (Correct Treatment)

Unknown vectors: $\begin{bmatrix} |V_1| & |V_2| & |V_3| & |V_4| & |V_5| \end{bmatrix}^T$ and $\begin{bmatrix} \delta_1 & \delta_2 & \delta_3 & \delta_4 & \delta_5 \end{bmatrix}^T$.

Bus 1: Gen → slack / ref. $|V_1| = |V_{1,set}|,\ \delta_1 = 0$

Bus 2: Load bus → PQ $P_2 = P_{2,Gen} - P_{2,Load},\ Q_2 = Q_{2,Gen} - Q_{2,Load}$

Bus 3: Gen bus → PV $P_3 = P_{Gen3} - P_{Load3},\ |V_3| = |V_{3,set}|$

Bus 4: Load bus $P_4 = P_{4,Gen} - P_{4,Load},\ Q_4 = Q_{4,Gen} - Q_{4,Load}$

Bus 5: Load bus $P_5,\ Q_5$

Step 1) Read data, Y_{Bus}, assume initial voltages and angles:

$$\begin{bmatrix} |V_1| \\ |V_2| \\ |V_3| \\ |V_4| \\ |V_5| \end{bmatrix} = \begin{bmatrix} |V_{1,set}| \\ 1.0 \\ |V_{3,set}| \\ 1.0 \\ 1.0 \end{bmatrix}, \qquad \begin{bmatrix} \delta_1 \\ \delta_2 \\ \vdots \\ \delta_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \qquad k = 0$$

Step 2) Compute V_i^{k+1}:

- V_1: slack, not updated $\Rightarrow$ $V_1 = |V_{1,set}|\angle 0°$
- V_2: from the update equation $\Rightarrow$ $|V_2|\angle\delta_2$
- V_3: PV bus $\Rightarrow$ compute $Q_{3,cal}$.
  - If $Q_{3,\min} \le Q_3 \le Q_{3,\max}$: compute $V_3 = |V_3|\angle\delta_3$, then reset magnitude: $V_3 = |V_{3,set}|\angle\delta_3$.
  - If any limit violation is there: $V_3 = |V_3|\angle\delta_3$ (bus treated as PQ with Q_3 fixed at the limit).
- 4th and 5th $\rightarrow$ load / PQ: $V_4 = |V_4|\angle\delta_4$, $V_5 = |V_5|\angle\delta_5$

False PV bus switching

A PV bus violating its Q-limit in one iteration and then switching back to its previous status (PV) in a later iteration.

Step 3) Check for convergence. If not converged, then repeat from Step 2.

6 Newton–Raphson Formulation

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In each iteration, we update $|V|$ and δ, where $V_i = |V_i|\angle\delta_i$.

For different types of buses, we have different known variables.

Load bus: P_i, Q_i known (scheduling):

$$P_i = P_{Gen,i} - P_{Load,i}, \qquad Q_i = Q_{Gen,i} - Q_{Load,i}$$

$$P_i - jQ_i = V_i^* \sum_{j=1}^{n} Y_{ij} V_j$$

What is known?

- $Y_{Bus} \Rightarrow$ known
- $|V_i|, \delta_i \Rightarrow$? (unknown)
- $P_{i,sp} \rightarrow$ from specification $\leftarrow$ balancing (from tertiary-level control)
- $Q_{i,sp} \rightarrow$ from specification $\leftarrow$ balancing
- $P_{i,cal}$, $Q_{i,cal} \Rightarrow$ using Y_{Bus} and $|V_i|, \delta_i$, $i = 1, 2, \ldots, n$

Generator bus: $P_i \Rightarrow P_{i,sp}$ and $P_{i,cal}$.

Load bus: $P_i \Rightarrow P_{i,sp}$ and $P_{i,cal}$; $\quad Q_i \Rightarrow Q_{i,sp}$ and $Q_{i,cal}$.

6.1 Mismatch Vector

For an n-bus system:

$$\begin{bmatrix}\Delta P_1 & \Delta P_2 & \cdots & \Delta P_n & \Delta Q_1 & \Delta Q_2 & \cdots & \Delta Q_n\end{bmatrix}^T$$

Rows belonging to the slack bus ($\Delta P_1, \Delta Q_1$) and to PV buses (ΔQ) are removed.

Example – 5-bus system: Bus 1 – slack; Bus 2 – PV / Gen; Buses 3, 4, 5 – load.

$$\Delta P = \begin{bmatrix}\Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \\ \Delta P_5\end{bmatrix}, \qquad \Delta Q = \begin{bmatrix}\Delta Q_3 \\ \Delta Q_4 \\ \Delta Q_5\end{bmatrix}$$

$$\max\ \text{abs}\left(\begin{bmatrix}\Delta P \\ \Delta Q\end{bmatrix}\right) \leq 0.0001 \ \Rightarrow\ \textbf{stop}$$

6.2 Real and Reactive Power in Polar Form

With $Y = G + jB$, $V_i = |V_i|\angle\delta_i$, $Y_{ij} = G_{ij} + jB_{ij}$:

$$P_i - jQ_i = |V_i|\angle{-\delta_i} \sum_{j=1}^{n} (G_{ij} + jB_{ij})|V_j|\angle\delta_j$$

$$P_i - jQ_i = \sum_{j=1}^{n} |V_i||V_j|\angle(\delta_j - \delta_i)\,(G_{ij} + jB_{ij})$$

$$P_i - jQ_i = |V_i| \sum_{j=1}^{n} |V_j|\,(\cos\delta_{ji} + j\sin\delta_{ji})(G_{ij} + jB_{ij}) \tag{B}$$

where $\delta_{ji} = -\delta_{ij}$ $(\delta_{ij} = \delta_i - \delta_j)$.

Expanding (B) and separating real and imaginary parts:

$$P_i = |V_i| \sum_{j=1}^{n} |V_j|\,(G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij})$$

$$Q_i = |V_i| \sum_{j=1}^{n} |V_j|\,(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij})$$

(II)

6.3 Newton–Raphson Principle

Write the equations as $f(x) = 0$ (e.g. $P_i - |V_i|\sum\cdots = 0$). Taylor expansion about x^0:

$$f(x) = f(x^0) + \left.\frac{\partial f(x)}{\partial x_j}\right|_{x^0} \Delta x_j + \text{H.O.T.}, \qquad j = 1, 2, \ldots, n$$

$$\underbrace{f(x) - f(x^0)}_{\Delta f(x)} = \underbrace{\left.\frac{\partial f(x)}{\partial x}\right|_{x=x^0}}_{J\ \text{(Jacobian)}} \Delta x$$

$$\underbrace{\Delta x}_{\text{update}} = \left[J^{-1}\right] \underbrace{[\Delta f(x)]}_{\text{mismatch}}$$

State and mismatch vectors (5-bus: $2n \times 1 = 10 \times 1$), with $V_i = |V_i|\angle\delta_i$:

$$x = \begin{bmatrix} \delta \\ |V| \end{bmatrix} = \begin{bmatrix} \delta_1 & \cdots & \delta_5 & |V_1| & \cdots & |V_5| \end{bmatrix}^T, \qquad \Delta f(x) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} P_{sp} - P_{cal} \\ Q_{sp} - Q_{cal} \end{bmatrix}$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} \Delta P_1 & \cdots & \Delta P_5 & \Delta Q_1 & \cdots & \Delta Q_5 \end{bmatrix}^T$$

6.4 Jacobian

$$J = \begin{bmatrix} \dfrac{\partial P}{\partial \delta} & \dfrac{\partial P}{\partial |V|} \\ \dfrac{\partial Q}{\partial \delta} & \dfrac{\partial Q}{\partial |V|} \end{bmatrix}_{2n \times 2n \ (10\times 10)}$$

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = [J] \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix} \qquad \Longrightarrow \qquad \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix} = [J]^{-1} \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}$$

7 Newton–Raphson Method of Power Flow Analysis – Algorithm

1) Read data. Form Y_{Bus}. Assume voltages (magnitude and angle):

$$[V] = 1.0, \quad [\delta] = 0.0 \quad \Rightarrow \text{flat start}; \qquad k = 1 \text{ (iteration count)}$$

2) Calculate the mismatch vector and check for convergence:

$$\max \ \text{abs} \left(\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}^k \right) < 0.0001 \ \Rightarrow \ \textbf{stop}; \quad \text{else go to step 3.}$$

3) Calculate the Jacobian (at $[\delta, \ |V|]^k$) and invert it:

$$\begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}^k = \left[J^{-1}\right]^k \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}^k$$

4) Update:

$$\begin{bmatrix} \delta \\ |V| \end{bmatrix}^{k+1} = \begin{bmatrix} \delta \\ |V| \end{bmatrix}^k + \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}^k, \qquad k \Rightarrow k+1$$

Go back to step 2.

8 Jacobian Structure – 5-Bus Example

Bus 1 – Ref./slack ($|V_1|$, $\delta_1 = 0$ known); Bus 3 – PV bus / Gen (P_3, $|V_3|$ known; Q_3 not known); others – load / PQ.

$$\begin{bmatrix}\Delta P_1\\ \Delta P_2\\ \Delta P_3\\ \Delta P_4\\ \Delta P_5\\ \Delta Q_1\\ \Delta Q_2\\ \Delta Q_3\\ \Delta Q_4\\ \Delta Q_5\end{bmatrix} = \left[\begin{array}{c|c} J_1 = \dfrac{\partial P}{\partial \delta}\ (5\times 5) & J_2 = \dfrac{\partial P}{\partial |V|}\ (5\times 5) \\ \hline J_3 = \dfrac{\partial Q}{\partial \delta}\ (5\times 5) & J_4 = \dfrac{\partial Q}{\partial |V|}\ (5\times 5) \end{array}\right]_{10\times 10} \begin{bmatrix}\Delta\delta_1\\ \Delta\delta_2\\ \Delta\delta_3\\ \Delta\delta_4\\ \Delta\delta_5\\ \Delta|V_1|\\ \Delta|V_2|\\ \Delta|V_3|\\ \Delta|V_4|\\ \Delta|V_5|\end{bmatrix}$$

- $\Delta P_1 = P_{1,sp} - P_{1,cal}$
  - At the slack bus, P_{sp} and Q_{sp} are not available.
  - At the generator bus, Q_{sp} is not known.
- $\Delta\delta_1 = 0,\ \ \Delta|V_1| = 0,\ \ \Delta|V_3| = 0$

Full element form:

$$J = \left[\begin{array}{ccccc|ccccc} \frac{\partial P_1}{\partial \delta_1} & \frac{\partial P_1}{\partial \delta_2} & \frac{\partial P_1}{\partial \delta_3} & \frac{\partial P_1}{\partial \delta_4} & \frac{\partial P_1}{\partial \delta_5} & \frac{\partial P_1}{\partial |V_1|} & \frac{\partial P_1}{\partial |V_2|} & \frac{\partial P_1}{\partial |V_3|} & \frac{\partial P_1}{\partial |V_4|} & \frac{\partial P_1}{\partial |V_5|} \\ \frac{\partial P_2}{\partial \delta_1} & \frac{\partial P_2}{\partial \delta_2} & \cdots & & & \frac{\partial P_2}{\partial |V_1|} & \cdots & & & \\ \vdots & & \ddots & & & \vdots & & & & \\ \frac{\partial P_5}{\partial \delta_1} & & & & \frac{\partial P_5}{\partial \delta_5} & \frac{\partial P_5}{\partial |V_1|} & & & & \frac{\partial P_5}{\partial |V_5|} \\ \hline \frac{\partial Q_1}{\partial \delta_1} & \frac{\partial Q_1}{\partial \delta_2} & \cdots & & & \frac{\partial Q_1}{\partial |V_1|} & \frac{\partial Q_1}{\partial |V_2|} & \cdots & & \\ \vdots & & \ddots & & & \vdots & & \ddots & & \\ \frac{\partial Q_5}{\partial \delta_1} & & & & \frac{\partial Q_5}{\partial \delta_5} & \frac{\partial Q_5}{\partial |V_1|} & & & & \frac{\partial Q_5}{\partial |V_5|} \end{array}\right] = \begin{bmatrix} J_1 & J_2 \\ J_3 & J_4 \end{bmatrix}$$

8.1 Sub-matrices

Each sub-matrix has diagonal ($j = i$) and off-diagonal ($j = 1, 2, \ldots, n,\ j \neq i$) terms, $i = 1, 2, \ldots, n$:

	Diagonal	**Off-diagonal**
J_1	$\dfrac{\partial P_i}{\partial \delta_i}$	$\dfrac{\partial P_i}{\partial \delta_j}$
J_2	$\dfrac{\partial P_i}{\partial \lvert V_i\rvert}$	$\dfrac{\partial P_i}{\partial \lvert V_j\rvert}$
J_3	$\dfrac{\partial Q_i}{\partial \delta_i}$	$\dfrac{\partial Q_i}{\partial \delta_j}$
J_4	$\dfrac{\partial Q_i}{\partial \lvert V_i\rvert}$	$\dfrac{\partial Q_i}{\partial \lvert V_j\rvert}$

Starting from $P_i = |V_i| \sum_{j=1}^{n} |V_j| \left(G_{ij} \cos\delta_{ij} + B_{ij} \sin\delta_{ij}\right)$, the elements are obtained by differentiating with respect to δ_i (diagonal) and δ_j (off-diagonal).

Added for completeness

$$\frac{\partial P_i}{\partial \delta_i} = -Q_i - B_{ii}|V_i|^2 \qquad \frac{\partial P_i}{\partial \delta_j} = |V_i||V_j|\left(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}\right)$$

$$\frac{\partial P_i}{\partial |V_i|} = \frac{P_i}{|V_i|} + G_{ii}|V_i| \qquad \frac{\partial P_i}{\partial |V_j|} = |V_i|\left(G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij}\right)$$

$$\frac{\partial Q_i}{\partial \delta_i} = P_i - G_{ii}|V_i|^2 \qquad \frac{\partial Q_i}{\partial \delta_j} = -|V_i||V_j|\left(G_{ij}\cos\delta_{ij} + B_{ij}\sin\delta_{ij}\right)$$

$$\frac{\partial Q_i}{\partial |V_i|} = \frac{Q_i}{|V_i|} - B_{ii}|V_i| \qquad \frac{\partial Q_i}{\partial |V_j|} = |V_i|\left(G_{ij}\sin\delta_{ij} - B_{ij}\cos\delta_{ij}\right)$$

8.2 Reducing the System

- **Slack bus** $\rightarrow$ remove equations related to ΔP, ΔQ and variables δ and $|V|$.
- **Gen (PV) bus** $\rightarrow$ remove equations related to ΔQ and variable $|V|$.

With bus 1 slack and bus 3 PV:

$$\begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \\ \Delta P_5 \\ \Delta Q_2 \\ \Delta Q_4 \\ \Delta Q_5 \end{bmatrix} = \Big[\ J\ \Big]_{7\times 7} \begin{bmatrix} \Delta\delta_2 \\ \Delta\delta_3 \\ \Delta\delta_4 \\ \Delta\delta_5 \\ \Delta|V_2| \\ \Delta|V_4| \\ \Delta|V_5| \end{bmatrix}$$

and solve them iteratively.

8.3 Handling PV Buses

In step 3, check the Q-limit (as in the Gauss–Seidel treatment: if violated, fix Q at the limit and treat the bus as PQ).

9 Power System Security

Minimum security criteria: $N-1$

- $N \Rightarrow$ components: generators, transformers, transmission lines, large loads, etc.
- On the outage of any one component, transformer/line power flows and voltages must remain within permissible limits.